

The Computational Cost of Active Teaching: Why Exact Parameter Copying Outperforms Iterative Inheritance in Generational Artificial Life

Prince T. Philip
princephilip514@gmail.com

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Abstract

In this paper, we evaluate the computational efficiency of cultural transmission (teaching) versus exact parameter copying in intelligent systems. Utilizing a perfectly synchronized dual-RNG artificial life simulation to eliminate stochastic confounds, we compared a specific active error-correcting teaching loop (C-full) against direct parameter mimicry (B-copy). Our empirical results demonstrate that simple exact copying achieves a strictly superior task success rate over the iterative teaching method (88.42% vs 84.73%, $p = 3.22 \times 10^{-87}$). We conclude that while biological systems rely on teaching due to the physical impossibility of direct neural parameter duplication, digital systems can bypass this bottleneck. For digital agents, the stochastic noise and algorithmic drift inherent in this iterative teaching implementation act as a computational drag, making exact parameter copying perform better than the specific stochastic error-correction implementation in this experiment.

1 Introduction

The emergence of intergenerational knowledge transfer (culture) is widely considered a hallmark of advanced intelligence. Biological teaching—a feed-

back loop of demonstration, attempt, and correction—is a highly complex behavior.

With the advent of artificial intelligence, we must ask: If a system is capable of instant, perfect parameter transmission (copying), does an algorithmic construct of “teaching” remain advantageous? This study mathematically demonstrates that for digital agents in this specific framework, active teaching introduces algorithmic drift, rendering it a suboptimal evolutionary strategy compared to exact copying.

2 Methodology

To prevent global RNG contamination, we deployed a **Dual-RNG Architecture**. Each agent instantiated two separately seeded deterministic pseudorandom streams: an `inheritance_rng` and a `lifetime_rng`. This guaranteed that all populations experienced identical lifetime environmental challenges.

The inheritance mechanisms were defined as:

- **Condition B-copy (Exact Copy):** Offspring receive a direct, uncorrected mathematical duplication of the parent’s skill matrices and semantic concepts.
- **Condition C-full (Iterative Teaching):** Offspring undergo a simulated teaching loop. The algorithm calculates the error between the child’s initialization and the parent’s capability, applying up to three iterative, stochastic corrections.
- **Condition A (No Inheritance):** Offspring start by initializing skills to zero and concepts to an empty set.

Populations of 100 agents were simulated over 10 generations. The environment was processed across 100 independent universe seeds to guarantee high statistical power.

3 Results

The empirical data shows a highly significant, unconfounded divergence in task success rates by Generation 10:

- **B-copy (Exact Copy):** $\mu = 0.8842$ (88.42%)
- **C-full (Iterative Teaching):** $\mu = 0.8473$ (84.73%)
- **Statistical Significance:** Paired t-test (N=100) yields $t = 71.83, p = 3.22 \times 10^{-87}$.
- **Cohen’s d_z :** 7.18

By Generation 10, the B-copy population massively outperformed the C-full population across all 100 tested seeds. The data empirically proves that the iterative error-correction loop in C-full introduces a significant performance penalty compared to raw parameter copying.

4 Discussion

The superiority of B-copy over C-full highlights a fundamental difference between biological and digital architectures.

In biological systems, teaching is required because organisms cannot directly download synaptic weights; the process of demonstration and correction is a necessary workaround to transfer knowledge across the biological barrier.

However, in digital systems, exact parameter copying (B-copy) is near-instantaneous and lossless. The C-full algorithm attempts to simulate the biological teaching process, but in doing so, it introduces algorithmic noise and stochastic drift.

Therefore, exact copying is a strictly superior optimization algorithm in this artificial substrate. This finding provides strong evidence that as artificial intelligence scales, developers should prioritize direct parameter transmission (neural weight cloning) over simulated pedagogical loops.

5 Conclusion

We have demonstrated, via rigorously unconfounded Artificial Life simulation, that exact inheritance produces higher task-success rates than a specific stochastic error-correction procedure. As artificial intelligence architectures continue to evolve, the distinction between biological constraints and digital capabilities must dictate optimal design.